Module 2: Infrastructure Security

Topic 2: Network Security

Network Fundamentals

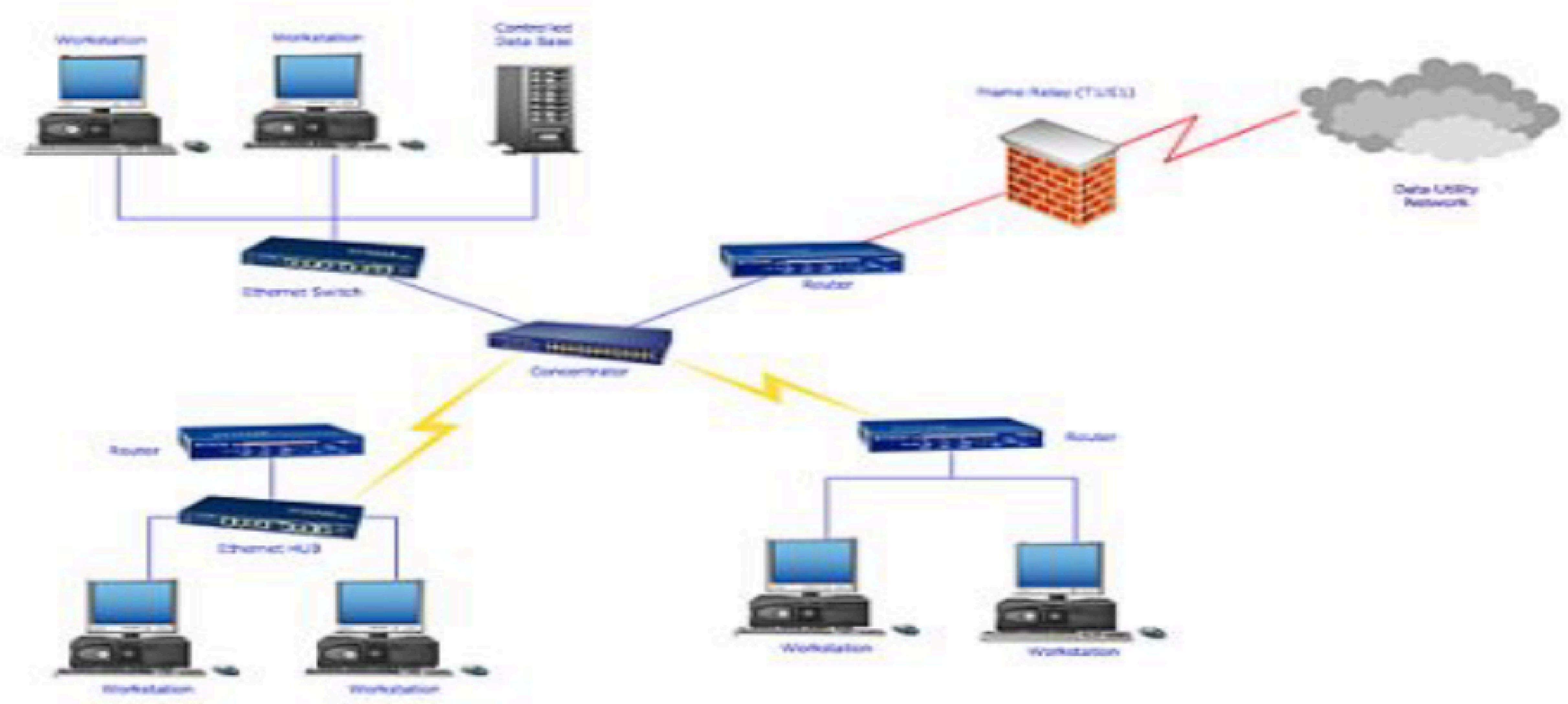
LAN Local Area Network: (local area network) is a group of computers and network devices connected together.

WAN Wide Area Network : A WAN connects several LANs, and may be limited to an enterprise (a corporation or an organization) or accessible to the public.

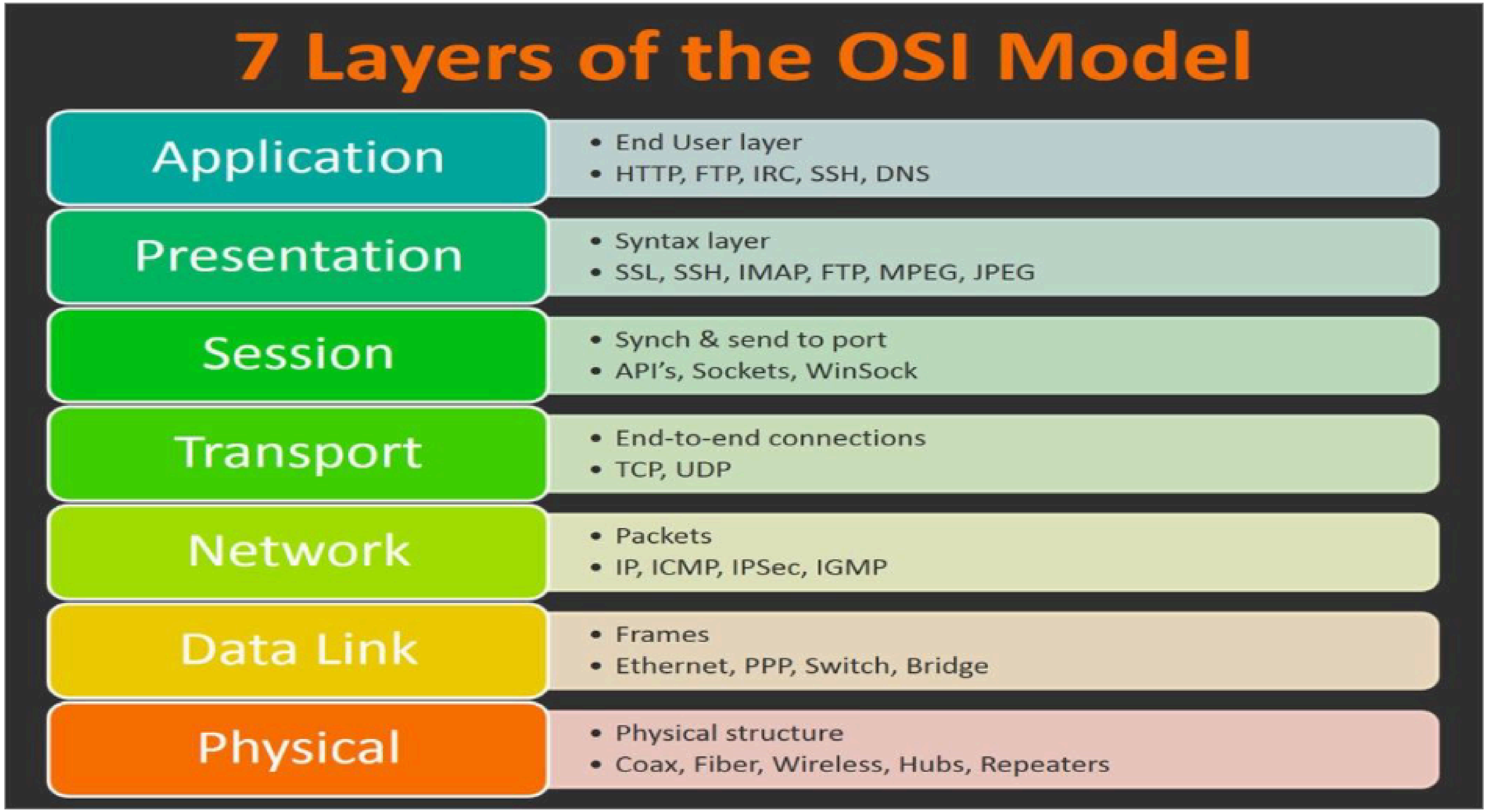
Network Fundamentals

LAN Local Area Network

WAN Wide Area Network



Network Fundamentals



Network Vulnerabilities

- **Outdated network systems (Routers, Switches, Firewalls, etc.)**

Exposes vulnerable system or entire network to attacks.

- **Misconfigured Network System**

Incorrect allow rules or default policies or credentials could help attacker in network attacks

- **Insecure Network Devices**

Using unknown brands or appliances known to have vulnerabilities or insecure in the network or putting device in open area location.

- **Insecure Wireless Networks**

Using obsolete security protocols such as WEP, WPS or using weak passwords.

Network Vulnerabilities

- **Network Sniffing**

Using advanced tools on the network to extract unauthorized information from the transferred traffic.

- **Man in the Middle**

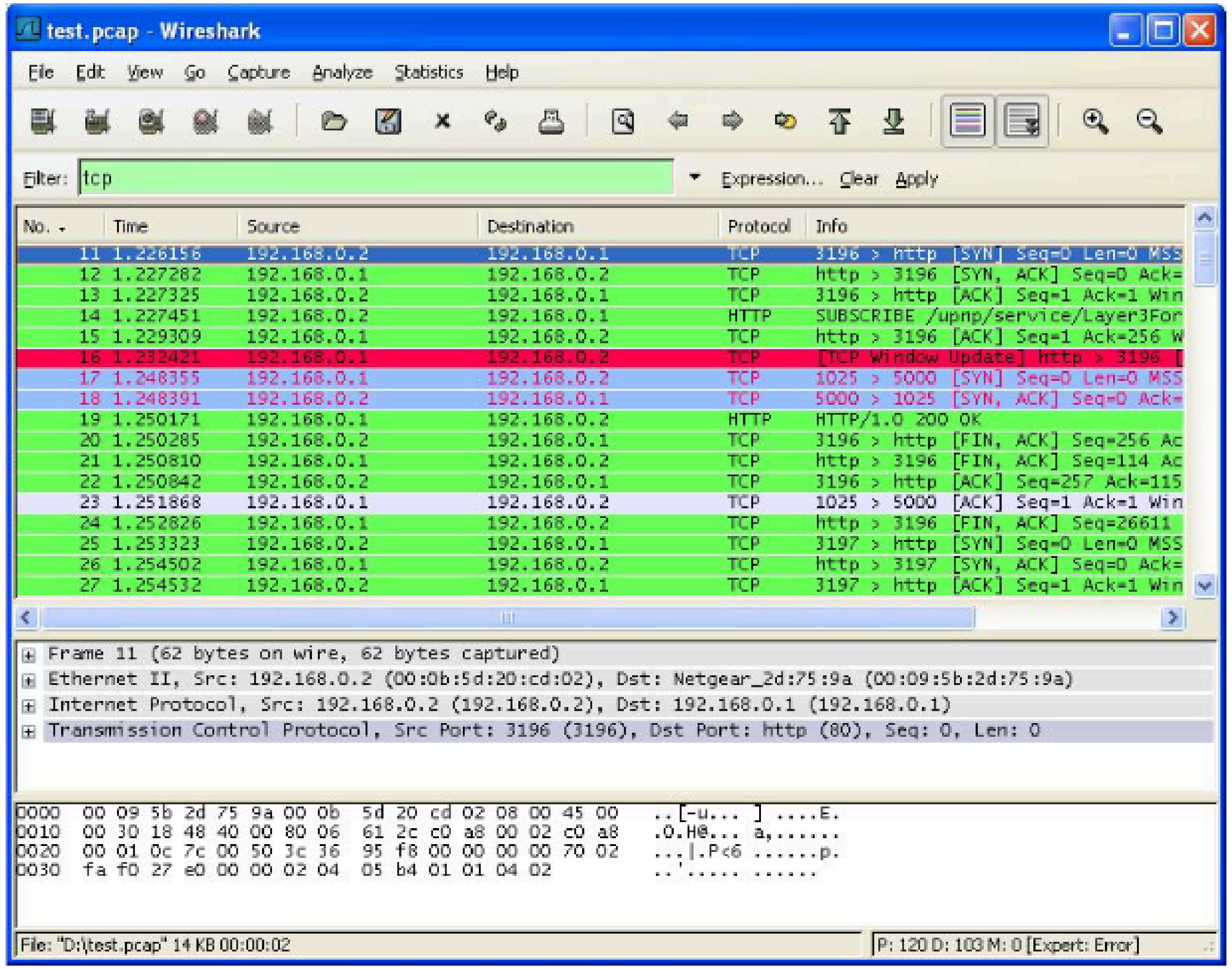
Attacker will try to intercept communication between 2 parties and change the content secretly without being noticed.

- **Shadow IT System**

Unauthorized system installed without approval of IT and Security teams.

Wireshark

- Widely used tool to analyze traffic.
- One of the skills that is important to security engineers to be able to understand traffic & troubleshoot security problems.
- Cross platform (Can work on both Windows & Linux OS)
- Can be used as well in sniffing attacks.



How to use Wireshark

- HTTP (Wireshark filter: tcp.port == 80)
  - In-the-clear web communications
- FTP/TFTP (Wireshark filter: tcp.port == 20 or tcp.port == 21)
  - File transfer without encryption
- Telnet (Wireshark filter: tcp.port == 23)
  - Remote login without encryption
- SMTP (tcp.port==25)/POP (tcp.port==110)/IMAP (tcp.port==143)
  - Email communication protocols

- Protocols to ignore (*unless there is a method provided to break encryption*)
  - HTTPS – encrypted web traffic
  - SSH – encrypted remote login
  - SFTP – secure (encrypted) file transfer
  - SMTP (port 465)/IMAPS (port 993)/POP (port 995) – secure email access

How to use Wireshark

Packets from 192.168.1.1	ip.src==192.168.1.1
Packets to and from port 80	tcp.port==80
From 10.10.3.2 to 10.10.3.40	ip.src==10.10.3.2 && ip.dst==10.10.3.40
To/from 10.10.3.2 on port 443	ip.addr==10.10.3.2 && tcp.port==443

How to Secure Network?

- Use firewall to protect against network attacks.

- Ensure correct design and architecture to the network.

- Use trusted security devices in the network if required (ex: IPS, Proxy, DDOS mitigator, etc).

- Ensure devices are up to date.

How to Secure Network?

- Ensure proper configurations are applied on the network devices.

- Enforce traffic encryption externally and internally.

- Use management tool to collect logs and errors from network devices.

- Use WPA2 with strong password or WPA2 enterprise and disable WPS on wireless network.